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# Timely Estimates of the Monthly Mexican Economic Activity

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In this article, we present a new approach based on dynamic factor models (DFMs) to perform accurate nowcasts for the percentage annual variation of the Mexican Global Economic Activity Indicator (IGAE), the commonly used variable as an approximation of monthly GDP. The procedure exploits the contemporaneous relationship of the timely traditional macroeconomic time series and nontraditional variables as Google Trends with respect to the IGAE. We evaluate the performance of the approach in a pseudo real-time framework, which includes the pandemic of COVID-19, and conclude that the procedure obtains accurate estimates, for one and two-steps ahead, above all, given the use of Google Trends. Another contribution for economic nowcasting is that the approach allows to disentangle the key variables in the DFM by estimating the confidence interval for the factor loadings, hence allows to evaluate the statistical significance of the variables in the DFM. This approach is used in official statistics to obtain preliminary and accurate estimates for IGAE up to 40 days before the official data release.

*Key words:* Dynamic factor models; global mexican economic activity indicator; google trends; nowcasts; pseudo real-time.

## 1. Introduction

Currently, the large amount of economic and financial time series collected over several years by official statistics agencies allows researchers to implement statistical and econometric methodologies that can generate accurate models to understand and anticipate macroeconomic phenomena. One of the most important events to anticipate are the movements of the gross domestic product (GDP) because they allow policy, and investment decisions to be made with greater certainty, in the expected scenario. For instance, if an economic contraction is foreseeable, businesses can adjust their expansion plans, governments can apply countercyclical policy, and consumers can adjust their spending patterns. In this context, "nowcasting" or timely estimates of macroeconomic time series is very relevant for policy makers and market participants to make informed decisions as a function of the short term of the economy. This is more evident for central

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banks, whose influential role during the pandemic has required them to adapt their communication policies to the incumbent state of the economy. For example, Benchimol et al. (2021) document such behavior using state-ot-the-art text-mining methodologies to analyze both pre and post pandemics communications. They find out that communications and actions have been more reactive to the COVID-19 crisis than to the global financial crisis and the dot-com crisis.

A joint econometric, statistical, and computational analysis with transcendental information is relevant to be able to tackle the complexity of generating accurate estimations for key variables of the economy, above all when unexpected phenomenon sharply affects the short run of the economy. Hence, this article proposes a novel approach to best exploit the available relevant information, which would allow to generate informative decisions for policy, decision making and is of interest for the society in general. We focus on Mexico's economy by using econometric models that are frequently used in the literature, but in our case, we estimate them by exploiting not only traditional information but also nontraditional variables, as has been recently used in other econometric applications.

In this sense, it is clear that as new economic and financial information is released, the estimates for a certain period are constantly being updated as well; thus, repetitive GDP estimations are made for the same quarter because unexpected economical or non-economical events can drastically affect the economy in the short term. Hence the need to use not only traditional economic and financial information, but also nontraditional high-frequency indicators, such as news, topics extracted from the Internet searches, social networks analysis, and so on.

The seminal work of Varian (2014) has become an obligatory reference for the inclusion of high-frequency information in econometric analysis. Buono et al. (2017) is also an important reference to characterize the types of nontraditional data and the econometric methods usually employed to extract information from the data. Both studies conclude that online search information is appropriate when the goal is to generate nowcasts of macroeconomic time series. More specific, Google Trends topics, an up-to-date source of online search information tool which provides an index of Internet searches or queries by category and geography, is frequently used to predict economic phenomena. See for example Stephens-Davidowitz and Varian (2014) which provides a full review of this tool, and other analytical tools from Google, as applied in the social sciences.

The inclusion of nontraditional information in prediction models has been well investigated in the literature, with evidence mostly in favor of its value in improving predictions. For example, Choi and Varian (2012), among others, have shown the effectiveness of Google Trends to improve predictions in real time for some United States (US) variables as motor vehicles and parts and unemployment benefits, the Hong Kong tourism and the Australian consumer confidence. Recent applications include Goldsmith-Pinkham and Sojourner (2020), who nowcast the number of workers filing unemployment insurance claims in the US, Caperna et al. (2022), who develop random forest models to nowcast country-level unemployment figures for the 27 European Union countries and Ali et al. (2021), who analyzes online job postings in the US childcare market under stay-at-home orders.

Note that the term nowcast is also relevant from an econometric and statistical standpoint, because we can use a rich variety of information to model, from a multivariate point of view, macroeconomic and financial events, plus specific incidents that can affect the dynamics of GDP in the short run. In this sense, these facts are related to the literature of dynamic factor models (DFMs), because a large amount of time series is useful to estimate underlying common factors.

In this order of ideas, we claim that Google search data, when combined with appropriate statistical tools has highly predictive power when applied in nowcasting models based on DFMs. In particular, we are interested in introducing Google search data along with DFMs to nowcast the monthly GDP proxy, officially published in Mexico by the Mexican National Institute of Statistics and Geography (INEGI) through the Global Economic Activity Indicator (IGAE), which covers approximately 95% of the total economy. It is released two months after the reference month (INEGI. a). Thus, by generating nowcasts for IGAE in an accurate and timely manner, the GDP's nowcasts can be drastically improved according to Gálvez-Soriano (2020).

Nowcasting the Mexican economy becomes relevant because of its importance in the world economy and in North America. According to data from the World Bank, in 2020 the Mexican GDP was ranked fifteenth in the world and fourth in the Americas. Moreover, since 1994 Mexico has been on a trade agreement with the US and Canada, which is currently known as the Agreement between the United States of America, the United Mexican States, and Canada, and formerly known as the North America Free Trade Agreement. Consequently, the development of new approaches to predict the Mexican economy activity is of relevance not only for Mexico but for at least the three countries.

DFM and other methodologies have been used to nowcast or predict the Mexican economy. To generate timely estimates, Delajara et al. (2016) use small-scale mixed-frequency DFM to nowcast, backcast and forecast GDP figures. Also, in one of the first works along this line Corona et al. (2017a), estimated common trends in a large and non-stationary DFM to predict the IGAE levels two steps ahead and concluded that error prediction was reduced with respect to some benchmarking univariate and multivariate time-series models. Caruso (2018) focuses on international indicators, mainly for the US economy, to show that its nowcasts of quarterly GDP outperform the predictions obtained by professional forecasters. Recently, Gálvez-Soriano (2020) concluded that bridge equations perform better than DFM and static principal components (PC) when making the nowcasts of quarterly GDP. It is worth to mention, because it is related to timely GDP estimation, Guerrero et al. (2013) in which, based on vector autoregression (VAR) models, they generate rapid GDP estimates (and its three grand economic activities) with a delay of up to 15 days from the end of the reference quarter, while the official GDP takes around 52 days after the quarter closes. Guerrero's work is the main reference for GDP flash estimates. "Estimación Oportuna del PIB Trimestral." (INEGI. b).

In our article, we deviate from prior studies relevant for Mexico, by including nontraditional information to capture more drastic frictions that occur in the very short run, in a time span of one or two months. We identified that previous works focus only on traditional information, which bounds their capacity to predict the recent historical declines attributed to COVID-19 and the associated economic closures since March 2020. A possible reason in the previous Mexican econometric analysis for not including nontraditional

information is the fact that traditional variables actually perform well when there are not unexpected events. Additionally, our econometric analysis focuses on obtaining nowcasts for the monthly Mexican economic activity and gives an interpretation to its results from an economic point of view by using timely macroeconomic and financial time series plus other nontraditional time series commonly used in big data analysis, specifically Google Trends.

We address the peculiarities previously mentioned for the Mexican economic activity by making use of previous knowledge to fill in the identified gaps by: (1) building a timely and correlated database using traditional economic and financial time series and real-time nontraditional information, determining the latter relevant variables with least absolute selection and shrinkage operator (LASSO) regression and other methods of time series selection variables; (2) estimating the common factors using the two-step (2SM) methodology of Doz et al. (2011) and (3) training and testing univariate time series models with the DFM's common factors to select the best nowcasts and combine the statistically equal better nowcasts to generate the current estimates. Additionally, to increase the interpretability of the results, we determine the confidence intervals for both the factor loadings and the factor itself to analyze the importance of each variable and the uncertainty attributed to the estimation.

By providing this new modeling strategy we consider the contributions of this article to be timeliness, openness and interpretability. First, given the timely availability of the information that our approach uses, we can generate nowcasts of the IGAE up to 40 days before the official data release; thus, our approach becomes an alternative to obtaining IGAE's preliminary estimates, which are very important in official statistics. Second, this article illustrates to practitioners the empirical strategy to generate IGAE nowcasts step-by-step, so that any user can replicate the results for other time series. Third, we can disentangle the contribution of each timely time series with respect to the economic activity, that is, the relative importance of traditional and nontraditional variables that encompasses the DFM. In this way, our objective is to generate nowcasts that yield an economic explanation of the nowcasts, while we do not pretend to carry out estimates with a pure statistical model or a non-interpretive approach. Consequently, we consider our approach to be both a predictive and an explicative model, in the sense of Shmueli (2010).

Another contribution of this article is to analyze and model the Mexican economic activity in pseudo real-time during the previous and current stage of the COVID-19 pandemic, by evaluating the performance of this approach with respect to that of other approaches. Furthermore, this article provides methodological support for the Mexican Indicador Oportuno de la Actividad Económica reported as experimental statistics by (INEGI. c).

The remainder of this article is structured as follows. Section 2 presents the methodology considered to generate the nowcasts. Section 3 describes the data and the descriptive analysis. Section 4 contains the empirical results. Finally, Section 5 concludes the article.

## 2. Methodology

This section describes how we employ DFM to generate the nowcasts of the IGAE. First, we describe how we build the correlated database by using traditional and nontraditional

time series. Then, we report how the stationary DFM shrinks the complete dataset in the 2SM strategy to obtain the estimated factor loadings and common factors and the Onatski (2010) procedure to detect the number of common factors. Finally, we describe the nowcasting approach.

## *2.1. Building the Database*

### 2.1.1. LASSO Regression: Selecting the Google Topics

The goal of using LASSO regression in our research is to select Google topics with most predictive power with respect to IGAE because, we do not have *a priori* knowledge about which ones are the most relevant across time. In contrast, regarding traditional data there exists a vast literature that indicates which variables are related with the economic activity. Hence, we propose an iterative algorithm described in detail in this subsection.

LASSO regression was introduced by Tibshirani (1996) as a new method of estimation, in linear models, that minimizes the residual sum of the squares (RSS) subject to the sum of the absolute value of the coefficients being less than a constant. In macroeconomic applications, Aprigliano and Bencivelli (2013) use the LASSO regression to select the relevant economic and financial variables in a large data set with the goal of estimating a new Italian coincident indicator.

Let $W_t = (w_{1t}, \ldots, w_{Kt})'$ be a $K \times 1$ vector of stationary and standardized variables. Consider the following penalized RSS:

$$\min_{RSS} = (y - W\beta)'(y - W\beta) + \lambda \sum_{j=1}^{K} |\beta_j|, \tag{1}$$

where $y = (y_1, \ldots, y_T)'$ is a $T \times 1$ vector, $\beta = (\beta_1, \ldots \beta_K)'$ is a $K \times 1$ vector, $W = (W_1, \ldots, W_T)'$ is a $T \times K$ matrix and $\lambda$ is a tuning parameter that controls the shrinkage of the estimates.

Tibshirani (1996) considers that this problem is equivalent to minimizing the sum of squares subject to $f(\beta) = \sum_{j=1}^{K} |\beta_j| \leq c$, where $c$ is a predetermined parameter that indicates the degree of regularization; in this case, the solution of Equation (1) is not closed, and it is obtained by convex optimization techniques. The LASSO solution has the following implications: (a) when $\lambda \rightarrow 0$, we obtain solutions similar to OLS, and (b) when $\lambda \rightarrow \infty$, $\hat{\beta}_{\lambda}^{LASSO} \rightarrow 0$. Therefore, the LASSO regression can perform as a variable selection method in linear models. Consequently, if $\lambda$ is large, more coefficients tend to zero, selecting the variables that minimize the error prediction.

LASSO regression is related to Ridge regression, but the latter focuses on determining the tuning parameter, $\lambda$, that controls the regularization effect. Where $f(b) = \sum_{j=1}^{K} |\beta_j^2$, solution is $\hat{\beta}_{\lambda}^{Ridge} = \left(W'W - \lambda I_p\right)^{-1} W'y$. In practice, this solution never sets the coefficients to exactly zero; therefore, ridge regression cannot perform as a variable selection method in linear models, although its prediction ability is better than OLS.

Hence, for a sample $K$ topics on Google Trends, the relevant topics $l = 0, \ldots, \zeta$, with $\zeta \geq 0$ are selected with the LASSO regression as follows:

1. Split the data for $t = 1, \ldots, T^* - H_g$, where $H_g$ is the number of the last months equivalent to the sample size in the CV.

2. For $h = 1$ and for the sample of size $T^* - H_g + h$, estimate $\hat{\beta}_{\lambda,h}^{LASSO}$ in a grid between $10^{10}$ and $10^{-10}$, that is, the rank of possible values of $\lambda$ by using a fixed window, two periods in test set sample and a training percentage of 0.75. Compute the following vector of indicator variables:

$$\hat{\beta}_{j,h} = \begin{cases} 1 & \text{if } \hat{\beta}_{\lambda,h}^{LASSO} \neq 0 \\ 0 & \text{if } \hat{\beta}_{\lambda,h}^{LASSO} = 0 \end{cases}$$

3. Repeat 2 until $H_g$.
4. Define the $H_g \times K$ matrix, $\hat{\beta} = (\hat{\beta}_1, \ldots, \hat{\beta}_K)$, where $\hat{\beta}_j = (\hat{\beta}_{j,1}, \ldots, \hat{\beta}_{j,Hg})'$ is an $H_g \times 1$ vector.
5. Select the $l$ significant variables that satisfy the condition $\hat{\beta}_l = (\hat{\beta}_{l \in j} | 1\hat{\beta} > \varphi)$, where $\varphi$ is the $1-\alpha$ sample quantile of $1\hat{\beta}$ with $\mathbf{1}$ being a vector $1 \times H_g$ of ones.

With this procedure, we select the topics that frequently reduce the prediction error –out-of-sample– for the IGAE estimates during the last $H_g$ months. We estimate the optimum A by using the `glmnet` package, from the R program, while the CV in the LASSO procedure is implemented in the `caret` package.

Once the $\zeta$ topics have been selected, we can refine our search by estimating DFMs with them and the macroeconomic and financial variables with them. The intention is to select the combination of topics $\binom{\zeta}{j}$ for $l = 0, \ldots, \zeta$, that minimizes the out-of-sample error estimation in a CV, when the obtained common factor is included in a regression model. This procedure is computationally expensive but can guide us to the most relevant Google topics for our nowcasting objectives in an empirical way.

Specifically, in each $h = 1, \ldots, H_g$, we selected the most relevant topics through the following heuristic algorithm:

1. Set $h = 1$.
2. Selecting $\zeta$ relevant topics, consider the different combination of matrices $\binom{\zeta}{j}$ for $l = 0, \ldots, \zeta$. For example, for $l = 1$ we have $\varsigma$ matrices. The case of $l = 0$ is when we do not consider Google topics.
3. By using also the traditional information, we estimate dynamic factors for each combination of topics and they are introduced in linear regressions estimating the MAEs for the recent past months $h* = 1, \ldots, H_g^*$.
4. We select the combination -or matrix-that consider the topics that minimize the MAE.
5. Repeat the procedure until $h = H_g$.

This procedure allows us to obtain the relevant topics in the past for each $h$, consequently, this is essentially as doing nowcasts in pseudo real time and not only for the incumbent period.

### 2.1.2. Transformations

We propose to carry out transformations on the variables of the DFM to orient the estimated common factors so that they are highly correlated contemporaneously with the IGAE, so that the estimated common factors are more appropriate to our nowcasting goals.

Therefore, we define the annual percentage variation of IGAE as $y^* = (y_1, \ldots, y_{T^*})$, where $T^* = T-2$; hence, to predict $y^*$, the macroeconomic, financial and nontraditional time series are collected in $X_i^* = (x_{i1}^*, \ldots, x_{iT^*}^*)$, and are transformed such that they satisfy the following condition:

$$X_i = \left(f\left(X_i^*\right)\Big|_{corr}^{\max}\left(f\left(X_i^*\right), y^*\right)\right). \tag{2}$$

Hence, we select the $f(X_i^*)$ that maximizes the correlation with $y^*$. Consider $f(\cdot)$ as follows:

1. None (n)
2. Monthly percentage variation (m): $\left(\frac{X_t^*}{X_{t-1}^*} \times 100\right) - 100$
3. Annual percentage variation (a): $\left(\frac{X_t^*}{X_{t-12}^*} \times 100\right) - 100$
4. Lagged (l): $X_{t-1}^*$

Consequently, each time series is transformed to have maximum correlation with the target time series.

### 2.2. *Dynamic Factor Model*

We use and specify a DFM to jointly summarize, the dynamics of the traditional and nontraditional time series by estimating the common factors and their contributions. Once estimated, the common factors are used in statistical prediction models to generate the timely estimates of the IGAE.

Hence, we consider a stationary DFM where the observations, $X_t$, are generated by the following process:

$$X_t = PF_t + \varepsilon_t, \tag{3}$$

$$\Phi(L)F_t = \eta_t, \tag{4}$$

$$\Gamma(L)\varepsilon_t = a_t, \tag{5}$$

where $X_t = \left(x_{1t}, \ldots, x_{Nt}\right)'$ and $\varepsilon_t = (\varepsilon_{1t}, \ldots, \varepsilon_{Nt})'$ are $N \times 1$ vectors of the variables and idiosyncratic noises observed at time $t$. The common factors, $F_t = (F_{1t}, \ldots, F_{rt})'$, and the factor disturbances, $\eta_t = (\eta_{1t}, \ldots, \eta_{rt})'$, are $r \times 1$ vectors, with $r(r < N)$ being the number of static common factors, which is assumed to be known. The $N \times 1$ vector of idiosyncratic disturbances, $a_t$, is distributed independently of the factor disturbances, $\eta_t$, for all leads and lags, denoted by $L$, where $LX_t = X_{t-1}$. Furthermore, $\eta_t$ and $a_t$, are assumed to be Gaussian white noises with positive definite covariance matrices $\Sigma_\eta = \text{diag}(\sigma_{\eta 1}^2, \ldots \sigma_{\eta r}^2)$ and $\Sigma_a$, respectively. $P = (p_1, \ldots, p_N)'$, is the $N \times r$ matrix of factor loadings, where, $p_i = (p_{i1}, \ldots, p_{ir})'$ is an $r \times 1$ vector. Finally, $\Phi(L) = \mathrm{I} - \sum_{i=1}^{k} \Phi L^i$ and $\Gamma = I - \sum_{j=1}^{s} \Gamma L^j$, where $\Phi$ and $\Gamma$ are $r \times r$ and $N \times N$ matrices containing the VAR parameters of the factors and idiosyncratic components with $k$ and $s$ orders, respectively. For simplicity, we assume that the number of dynamic factors, $r_1$, is equal to $r$.

Alternative representations in the stationary case are given by Doz et al. (2011, 2012), who assume that $r$ can be different from $r_1$. Additionally, when $r = r_1$, Bai and Ng (2004), Choi (2017) and Corona et al. (2020) also assume possible non-stationarity in the idiosyncratic noises. Barigozzi et al. (2015, 2016) assume possible non-stationarity in $F_t$, $\varepsilon_t$ and $r \neq r_1$. Certainly, the literature to represent DFMs is vast. For example, Bai and Ng

(2002) and Bai (2004) focus on static ones, while Bai and Ng (2007) show the relationship between the static and the dynamic representation, in turn, Stock and Watson (2011) describe the dynamic model showing some estimation methods, including the PC method used in this work, and Forni et al. (2000) show the generalized DFM, where they make the static representation widely applicable. In our work, the dynamics of the common factors is specified as a VAR(p) model to refine the static estimation in a second step, as proposed by Doz et al. (2011), since it is important to assume that the idiosyncratic errors are stationary to obtain factor consistent estimates.

The DFM in Equations (3) to (5) is not identified. As we noted in the Introduction, the factor extraction used in this work is the 2SM; consequently, in the first step, we estimate the common factors by using PC to solve the identification problem and uniquely define the factors; we impose the restrictions $P'P/N = I_r$ and $F'F$ being diagonal, where $F = (F_1, \ldots, F_T)$ is $r \times T$. For a review of restrictions in the context of PC factor extraction, see Bai and Ng (2013).

Note that the transformations considered in this work do not have the goal of achieving stationarity as is frequently done in several empirical applications. Although intrinsically these transformations are stationary transformations regardless of whether $y*$ is stationary; in fact, the transformations $m$ and $\alpha$ tend to be stationary transformations when the time series are $I(1)$, which is frequent in economics; see Corona et al. (2017b). Otherwise, it is necessary that $(f(X_i^*), y^*)$ are cointegrated in the context of Bai (2004). This is so because even though, some common factor, $F_t$, can be non-stationary, consistent estimations remain even when the idiosyncratic errors are stationary. In this way, we use the PANIC test (Bai and Ng 2004) to verify this assumption. Note that, for example, Bai and Ng (2004) and Barigozzi et al. (2016) obtain consistent estimates for $F_t$ regardless of whether the idiosyncratic errors are non-stationary.

#### 2.2.1. Two-Step Method for Factor Extraction

Giannone et al. (2008) popularized the usage of 2SM factor extraction to estimate the common factors by using monthly information with the goal of generating the nowcasts of quarterly GDP. However, Doz et al. (2011) proved the statistical consistency of the estimated common factor using 2SM. In the first step, PC factor extraction consistently estimates the static common factors without assuming any distributions, allowing weak serial and cross-sectional correlation in the idiosyncratic noises; see, for example, Bai (2003). In the second step, we model the dynamics of the common factors via the Kalman smoother, allowing idiosyncratic heteroskedasticity, a situation that occurs frequently in practice. In a finite sample study, Corona et al. (2020) show that with the 2SM of Doz et al. (2011) based on PC and Kalman smoothing, we can obtain closer estimates of the common factors under several data generating processes that can occur in empirical analysis, such as heteroskedasticity, serial and cross-sectional correlation in idiosyncratic noises. Additionally, following Giannone et al. (2008), this method is useful when the objective is nowcasting given the flexibility to estimate common factors, when all variables are not updated at the same time.

The 2SM procedure is implemented according to the following steps:

1. Set $\hat{P}$ as $\sqrt{N}$ times the eigenvectors corresponding to the $r$ largest eigenvalues of $X'X$, where $X = (X_1, \ldots, X_T)'$ is a $T \times N$ matrix. By regressing $X$ on $\hat{P}$ and using the identifiability restrictions, obtain $\hat{F} = X\hat{P}/N$ and $\hat{\varepsilon} = X - \hat{F}'\,\hat{P}'$. Then, compute the asymptotic confidence intervals for both factor loadings and common factors as proposed by Bai (2003).
2. Set the estimated covariance matrix of the idiosyncratic errors as $\hat{\Psi} = \text{diag}\big(\hat{\sum}_{\varepsilon}\big)$, where the diagonal of $\hat{\Psi}$ includes the variances of each variable of $X$; hence, $\hat{\sigma}_1^2$ for $i = 1, \ldots, N$.
3. Estimate a VAR(k) model by OLS to the estimated common factors, $\hat{F}$, re express the autoregressive coefficients as the VAR(1) model, denoted by $\hat{\Phi}$. Assuming that $f_0 \sim N(0, \Sigma_f)$, the unconditional covariance matrix of the factors can be estimated as $\text{vec}\,(\hat{\sum}_f) = (I_{r2} - \hat{\Phi} \otimes \hat{\Phi})^{-1}\,\text{vec}\,(\hat{\sum}_{\eta})$, where $\hat{\sum}_{\eta} = \hat{\eta}'\hat{\eta}/T$.
4. Write DFM in Equations (3) to (5) in state-space form, and with the system matrices $\hat{P}$,$\hat{\Psi}$, $\hat{\Phi}$, $\hat{\sum}_{\eta}$ and $\hat{\sum}_f$, use the Kalman smoother to obtain an updated estimation of the factors denoted by $\tilde{F}$.

In practice, $X_t$ are not updated for all $t$; in these cases, we apply the Kalman smoother, $\tilde{F} = E(F_t|\Omega_T)$, where $\Omega_T$ is all the available information in the sample, and we take into account the following two cases:

$$\hat{\Psi}_i = \begin{cases} \hat{\sigma}_i^2 & \text{if } x_{it} \text{ is available,} \\ \infty & \text{if } x_{it} \text{ is not available.} \end{cases}$$

Empirically, when specific data on $X_t$ are not available, Harvey and Phillips (1979) suggests using a diffuse value equal to $10^7$; however, we use $10^{32}$ according to the package nowcast of the R program, see De Valk et al. (2019).

As commented in Stock and Watson (2011), the estimation of common factors based on PC factor extraction is consistent to the presence of breaks or time variation in the factor loadings. Additionally, in our case $X_t$ is seasonally adjusted, that is, we have removed outliers, and the time series selection and transformations is oriented to maximizing the contemporaneous relationship with IGAE. The purpose is to summarize the common movements, including the drastic drops of the economy, thus, all the time series tend to have the same variability. On top of that, the ADF tests tend not to reject the null hypothesis of structural change. In our case, not rejecting the null hypothesis in PANIC tests computed to the idiosyncratic errors could point out the presence of possible structural change.

### 2.2.2. Determining the Number of Common Factors

To detect the estimated number of common factors, $\hat{r}$, Onatski (2010) proposes a procedure to use when the proportion of the observed variance attributed to the factors is small relative to that attributed to the idiosyncratic term. This method determines a sharp threshold, which consistently separates the bounded and diverging eigenvalues of $X'X/T$. Note that if $X$ is centered and scaled, $X'X/T$ is the covariance matrix.

This algorithm is known as edge distribution, and Onatski (2010) provide the algorithm that generates a consistent estimator of $r$ for any fixed $\delta > 0$, denoted by $\hat{r}$. Corona et al.

(2017b) shows that this method works reasonably well in small samples. Two important features of this method are that the number of factors can be estimated without previously estimating the common components and that the common factors may be integrated.

### 2.3. Obtaining Nowcasts

The nowcasting models are basically regressions with ARMA errors, in which the dependent variable is the annual percentage variation of the IGAE and the exogenous variable are the estimated common factors. We select the autoregressive and moving average orders in a time series CV period, so that, they minimize the one-step ahead estimation error. In this subsection, we describe in detail the algorithm.

Having estimated the common factors as described in Subsection 2.2.1 by using $X_t$, we estimate a linear regression model with autoregressive moving average (ARMA) errors to generate the nowcasts

$$y_t^* = a + b\tilde{F}_t + u_t, t = 1, \ldots, T-2, \tag{6}$$

where $u_t = \gamma(L)\phi(L)^{-1}u_t$ with $\gamma(L) = 1 + \sum_{j=1}^{q}\gamma_j L^j$ and $\phi(L) = 1 - \sum_{i=1}^{p}\phi_i L^i$. The parameters are estimated by maximum likelihood. Consequently, the nowcasts are obtained using the following expression:

$$\hat{y}_{T^*+h} = \hat{a} + \hat{b}\tilde{F}_{T^*+h} + \hat{u}_{T^*+h} \text{ for } h = 1, 2. \tag{7}$$

Note that Giannone et al. (2008) propose using the model with $p = q = 0$; hence, the nowcasts are obtained by using expression (7). In our case, we estimate different models by the orders $p = 0, \ldots p_{\max}$ and $q = 0, \ldots q_{\max}$; thus, the case of Giannone et al. (2008) is a particular case of this expression. Our interest is in selecting models with similar performance in a CV period. In this way, we carry out the following procedure:

1. Start with $p = 0$ and $q = 0$.
2. Estimate the nowcasts for $T^* + 1$ and $T^* + 2$, namely, $\hat{y}^{0,0} = \left(\hat{y}_{T^*+1}, \hat{y}_{T^*+2}\right)'$.
3. Split the data for $t = 1, \ldots, T^* - H_t$.
4. For $h = 1$ and for the sample of size $T^* - H_t + h$, estimate Equation (6), generate the nowcasts with expression (7) one step ahead, and calculate the errors and absolute error (AE) as follows:

$$e_1^{0,0} = y_{T^*-H_t+1} - \hat{y}_{T^*-H_t+1}$$

$$AE_1^{0,0} = |e_1^{0,0}|$$

5. Repeat steps 3 and 4 until $H_t$. Then, estimate $e^{0,0} = \left(e_1^{0,0}, \ldots, e_H^{0,0}\right)'$ and $AE^{0,0} = \left(AE_1^{0,0}, \ldots, AE_{H_t}^{0,0}\right)$. Additionally, we define the weighted AE (WAE) as $WAE^{0,0} = AE^{0,0}\ \Upsilon$ where $\Upsilon$ is a weighted $H_t \times 1$ matrix that penalizes the nowcasting errors such that $\Upsilon 1' = 1$.
6. Repeat steps for all combinations of $p$ and $q$ until $p_{max}$ and $q_{max}$. Generate the following elements:

$$\hat{y}(p, q) = \left(\hat{y}^{0,0}, \hat{y}^{1,0}, \ldots, \hat{y}^{p_{\max}, q_{\max}}\right),$$

$$e(p,q) = \left(e^{0,0}, e^{1,0}, \ldots, e^{p_{\max},q_{\max}}\right),$$

$$WAE(p,q) = \left(WAE^{0,0}, WAE^{1,0}, \ldots, WAE^{p_{\max},q_{\max}}\right)',$$

where $\hat{y}$ is a $2 \times (p_{\max}+1)(q_{\max}+1)$ matrix of nowcasts, $e$ is an $H_t \times (p_{\max}+1) \times (q_{\max}+1)$ matrix that contains the nowcast errors in the data test, and $WAE$ is an $H_t \times 1$ vector of the weighted errors in data test.

7. We select the best nowcast as a function of $p$ and $q$, denoted by $\hat{y}(p*, q*)$, where $p*$, $q*$ are obtained as follows:

$$p^*, q^* = \underset{0 \le p,q \le p_{\max} q_{\max}}{\operatorname{argmin}} \; WAE(p,q)$$

8. To use models with similar performance, we combine the nowcasts of $\hat{y}(p*, q*)$ with models with equal forecast errors according to the Diebold and Mariano (1995) tests, by using the $e(p,q)$, carrying out pairs of tests between the model with minimum $AE(p,q)$ and the others. Consequently, from the models with statistically equal performance, we select the median of the nowcasts, namely, $\hat{y}$.

This nowcasting approach allows the generation of nowcasts based on a trained process, taking advantage of the information of similar models. In this case, $\hat{b}$ must be significant to exploit the relationship between the IGAE and the information summarized by the DFM. Note that $\Upsilon$ is a weighted matrix that penalizes the nowcasts errors. The most common form is $\Upsilon = (1/H_t, \ldots, 1/H_t)'$, a $H_t \times 1$ matrix where all nowcasts errors have equal weight named in literature as mean absolute error (MAE). Therefore, we are not considering by default the traditional MAE, but rather a weighted (or equal) average of the individual AE. For example, we could have penalized with more weight the last nowcasts errors, that is, in the COVID-19 period. Also, note that we can obtain $AE(p,q)$ and estimate the median or some specific quantile for each vector of this matrix.

Note that even though root mean squared errors (RMSEs) are often used in the forecast literature, we prefer a weighted function of AEs, although in this work we use equal weights that is, the MAE. The main advantages of MAE over RMSE are two: (1) it is easy to interpret since it represents the average deviation without considering its direction, while the RMSE averages the squared errors and then we apply the root, which tends to inflate the larger errors and (2) RMSE does not necessarily increase with the variance of the errors. Anyway, the two criteria are in the interval [0, ∞) and are indistinct to the sign of errors. In practical terms, in the rest of the work and whenever its appropriate, we use both indistinctly. This allows to compare our model's MAE with models whose error measure is the RMSE.

## 3. Data and Descriptive Analysis

To illustrate the behaviour of the target variable, Figure 1 shows the historical $y*$ from January 2004 to October 2020.

We can see that IGAE's fall in 2020 has been the sharpest in the recent history. The pandemic brought about -19.7%, -21.4% and -14.4% contractions in total economic




*Fig. 1. Evolution of the annual growth rate of IGAE*

activity for April, May and June of 2020, respectively, but the economy had already begun to show signs of deterioration since May 2019. Prior to the coronavirus outbreak, the only comparable economic turmoil was the economic contraction derived from the Great Recession of 2008–2009.

Between January 2019 and February 2020, before the COVID-19 outbreak started in Mexico, the annual growth of IGAE already showed signs of slowing and fluctuated around -1.75% and 76%, and since May 2019, the economy exhibited nine consecutive months of negative growth.

The first six months of the COVID-19 pandemic (until September 2020) has had severe impacts on the Mexican economy. The first case of coronavirus in Mexico was documented on February 27, 2020.

Broken down by sector, the economy suffered devastating the consequences in the secondary and tertiary sectors. The industrial sector registered the deepest contractions, reducing its activity in April and May by -29.9% and -29.6%, respectively, in annual terms, mainly driven by the closure of manufacturing and construction operations, which were considered nonessential businesses, following a slight recovery in June, -17.1%, when an important number of activities, including automobile manufacturing, resumed but remained at low activity levels. The services sector also suffered from lockdown measures, falling by -15.9%, -18.7% and -13.7% in the three months of the second quarter, respectively, especially on transportation, retail, lodging and food preparation, mainly due to the decrease in tourist activity, although restaurants and airports were not closed. The primary sector showed signs of resilience and even grew in April and May 2020, by 0.7% and 1.5%, and shrank in June by -2.4% on an annual basis.

On the other hand, note that the variables to estimate the DFM are selected by using the criteria of timely and contemporaneous correlation with respect to $y*$. Thus, we consider a select group of macroeconomic and financial time series that represent contemporaneously and timely, the industrial and service sectors. For example, we take into account

information related with the industrial production from Mexico and US, manufactures, exports, imports, sales, hotel occupation, number of formal workers, and so on. Note that usually the nowcasts models focus on predicting the quarterly GDP. In this sense, there are several monthly variables that anticipate the GDP release. Therefore, for quarterly time series nowcasting, the availability of information is larger.

In this sense, the model differs from those in the traditional literature on large DFMs, which use a large amount of economic and financial variables; see, for example, Corona et al. (2017a) who use 211 time series to estimate the DFM for the Mexican case with the goal of generating forecasts for the levels of IGAE. However, Gálvez-Soriano (2020) uses approximately 30 selected time series to generate nowcasts of Mexican quarterly GDP. Note that Boivin and Ng (2006), in the context of DFM, point out that we can reduce the forecast prediction error with selected variables by estimating the common components. Additionally, Poncela and Ruiz (2016) and Corona et al. (2020) show that with a relatively small sample size, for example, $N = 12$, we can accurately estimate a rotation of the common factors. We acknowledge that recent developments on the inclusion of big data using sparse or dense modeling to generate economic predictions, and have concluded that it is not always possible to identify sparse predictive representations that allow the selection of adequate predictors from a larger amount of macroeconomic variables (Giannone et al. 2021). Thus we keep with the economic variables selected a priori based on our literature review.

Alternatively, when the timeliness of the estimation becomes relevant the set of variables is restricted as well, for example, compared to Corona et al. (2017a) where the key variable is also monthly, although they do not require the timely criterion, thus they exploit the lag correlation between a large amount of time series with respect to IGAE's current value.

Consequently, given the timely and possibly contemporaneous correlation with respect to the $y*$, the variables considered in this work are described in Appendix (Section 6). All variables are seasonally adjusted in the following ways: (1) directly downloadable from their source or (2) by applying the X-13ARIMA-SEATS.

Hence, we initialized with 68 time series divided into traditional and nontraditional time series. The traditional time series is timely macroeconomic and financial information such as the Industrial Production Index values for Mexico and the US, business confidence, and exports, imports, Mexican stock market index, nominal exchange rate, interest rate and the Standard & Poor's 500. We considered the possibility of including other timely time series as inflation, oil prices, consumer confidence, among many others, but were discarded given that they inclusion in the DFM increased the error in the CV. Additionally, for the nontraditional variables, we have daily variables such as the media mobility index obtained from Twitter and the topics extracted from Google Trends. These topics are manually selected according to several phenomena that occur in Mexican society, such as frequencies of politicians' names, natural disasters, economic themes and topics related to COVID-19, such as coronavirus, quarantine, or face mask. The Google Trends variable takes a value of 0 when the topic is not searched in the time span and 100 when the topic has the maximum search in the time span. Although these variables are expressed as monthly variables, for the media mobility index, we average the daily values, and for Google Trends we download the variables by month. While searching for significant Google topics in the past, we normalize each Google topic as $\left(\frac{W_i}{max(W_i)}\right) \times 100$ to remove the effect of the maximum search on recent data.

The media mobility index is calculated based on Twitter information following Graff et al. (2022). We selected around 70,000 daily tweets georeferenced to the Mexican territory, each one is associated with a bounding box. Then, movement data analysis is performed by identifying users and their sequence of daily tweets: a trip is considered for each pair of consecutive geo-tagged tweets found in different bounding boxes. The total number of trips per day is obtained and divided by the average number of users in the month. The number obtained can be interpreted as the average number of trips that tweeters make per day. Twitter allows to estimate the index with a larger sample size than with respect other alternatives as Google or Facebook.

To select the relevant topics, we apply the methodology described in Subsection 2.1.1 by using $H_g = H_t + 1$ and $\alpha = 0.10$ where $H_t = 36$; consequently, we select the topics that are relevant in at least 90% of cases in a sample size that covers the number of months of the CV in the nowcasting approach and also, the period of the current nowcasts. Additionally, we select $H_i^* = 8$.

In this way, the current significant topics (as of January 15, 2021) are "health crisis", "quarantine", "face mask N95" and "recession". Other relevant topics in the past were "coronavirus", "dollar", "oil", "pandemic","unemployment", among many others.

Once $X^*$ is defined, we apply the transformations suggested by Equation (2) to define $X$. Appendix (Subsection 6.2. Table 1) shows each $X_i$ ordered according to its correlation with $y*$.

In order to highlight how the traditional and nontraditional variables are related with $y*$ in the last $H_g = 37$ months, Figure 2 shows the ordered linear correlations for different sample sizes $T^* - H_g + h$, with $h = 1, \ldots, 37$.

We can see that during the last 37 months the time series are strong or moderately correlated with the IGAE; the exceptions being SP 500 and REM, whose correlations have decreased starting the COVID-19 pandemic. Also, M4 is weakly correlated throughout the period. Although Google topics have become more correlated since COVID-19 started, in the past they were medium correlated, which implies its possible inclusion, not only in the current period but also in the past. Figure 2 also shows the selected topics when the selection is carried out with all sample size $T^*$, that is, when $h = 37$. Other interesting traditional variables are IRGS, HOTEL and GAS, which show a similar behavior as the Google topics. Intuitively, these variables are more relevant to explain the IGAE's movement since the

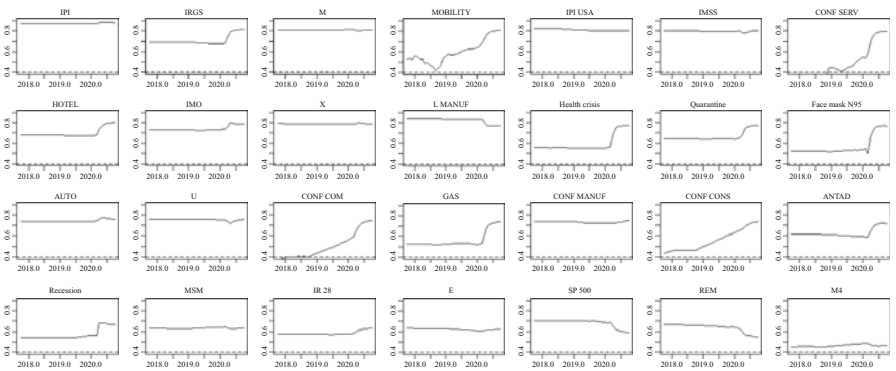



*Fig. 2. Linear correlation: between X and y* by using* $t = 1, \ldots, T* - H_g + h$ *with* $h = 1, \ldots, 37$.

COVID-19 pandemic started. Additionally, variables that exhibit linear correlations with a positive trend are MOBILITY and the those related to business confidence.

Note that the nowcasts depend on the dates of the information released. Depending on the day of the current month, we can obtain nowcasts with a larger or smaller percentage of updated variables. For example, the financial time series and nontraditional variables are available in real time, but macroeconomic traditional time series, which are timely with respect to the IGAE, are available on different moments according to their official release dates.

To obtain timely estimates, we rely on the fact that nontraditional variables, financial time series, business confidence and fuel demand, can be obtained on the day after the month $T^*$ closes. This indicates that on the first day of month $T^* + 1$, we can generate the nowcasts for $T^* + 2$ with approximately 50% of the updated information and 81% for the current month, $T^* + 1$. Moreover, in the 12th day, the IMSS variable is updated, and on the 16th, the IPI USA is usually updated. Consequently, in official statistics, we recommend conducting the nowcasts on the first day of $T^* + 1$ and 16 days later, updating the nowcasts with the latter two timely traditional and important time series, taking into account the timeliness of the most relevant variables. Note that IPI represents around the 34% of the monthly GDP, and represents more than 97% of the second grand economic activity. Given that the IPI is updated around ten days after the end of the reference month, this information is very valuable to carry out the $T^* + 1$ nowcasts.

Consequently, we carry out the nowcasts around the 15th day of each month. In the results presented herein, the database is current up to January 15, 2021; therefore, we generated the nowcasts ten days before the official result of November 2020 and 41 days before the official value of December 2020 was published, having 90% and 66% of updated variables at $T^* + 1$ and $T^* + 2$, respectively.

Appendix (Subsection 6.2, Table 2) shows the approximate day when the information is released for $T^* + 2$ after the current month $T^*$. Consequently, the temporal dimension of the panel to carry out the nowcasts for the last two months of 2020 is from January 2004 to December 2020. Whenever missing values exist before October 2020, they are imputed according to the methodology proposed by Giannone et al. (2008). Similarly, missing values for the nowcasts months are operated following the 2SM procedure described in Subsection 2.2.1.

## 4. Nowcasting Results

### *4.1. Estimating the Common Factors and the Loading Weights*

By applying the Onatski (2010) procedure to $X$ centered, we can conclude that $\hat{r} = 1$ is adequate to define the number of common factors. Alternative criteria as the ratio of eigenvalues of the sample covariance matrix of the observations (Ahn and Horenstein 2013) also conclude $\hat{r} = 1$. Note that the sample effective dependence in the correlation matrix $X$ is 0.79, which can be considered as relatively large. In other words, once the significant Google Trends topics are selected, the 28 time series observed for 204 months that compose the DFM are summarized with one latent variable. The estimated static common factor obtained by PC using the set of variables, $X$, their asymptotic confidence intervals at 95%, in Figure 3, the asymptotic variance of the estimated common factors is

calculated as $\frac{1}{N}\hat{V}^{-1}\Theta_t\hat{V}^{-1}$ where $\hat{V}$ is the $r \times r$ diagonal matrix of the first $r$ largest eigenvalues of matrix $X'X/(NT)$ and $\Theta_t = \frac{1}{N}\sum_i^N \hat{p}_i'\hat{p}_i'\hat{\varepsilon}_{it}^2$ and the dynamic factor estimates by applying the 2SM procedure with $k = 1$ lags.

We observe the common factors summarizing the previous elements representing the decline in the economy in 2009 and 2020. Note that in the last period, the dynamic common factor shows a slight recovery of the economy because this common factor supplies more timely information than the static common factor. Thus, the static common factor has information up to August 2020, while the dynamic factor has information up to October 2020. Note that the confidence intervals are closed with respect to the static common factor, which implies that the uncertainty attributed to the estimation is well modelled. Note that the dynamic factor can be interpreted as a coincident economic indicator because it measures contemporaneously the global state of the economy.

The correlation between $\tilde{F}_t$ and $y*$ is approximately 0.86 prior to the financial crisis of 2009, increasing from this year to 0.98, showing a slight decrease since 2011, dropping in 2016 to 0.95 and fully reaching levels of 0.96 since 2020. Consequently, we can exploit the contemporaneous relationship between the dynamic factor and the IGAE to generate their nowcasts for the two following months that the common factors have estimated with respect to the IGAE. Appendix (Subsection 6.2, Table 3) shows the correlation coefficient of $\tilde{F}_t$ with $y*$ since 2008.

Having estimated the dynamic factor by the 2SM approach, we show the results of the loading weight estimates that capture the specific contribution of the common factor to each variable, in other words, given the PC restrictions, they can be seen as $N$ times the

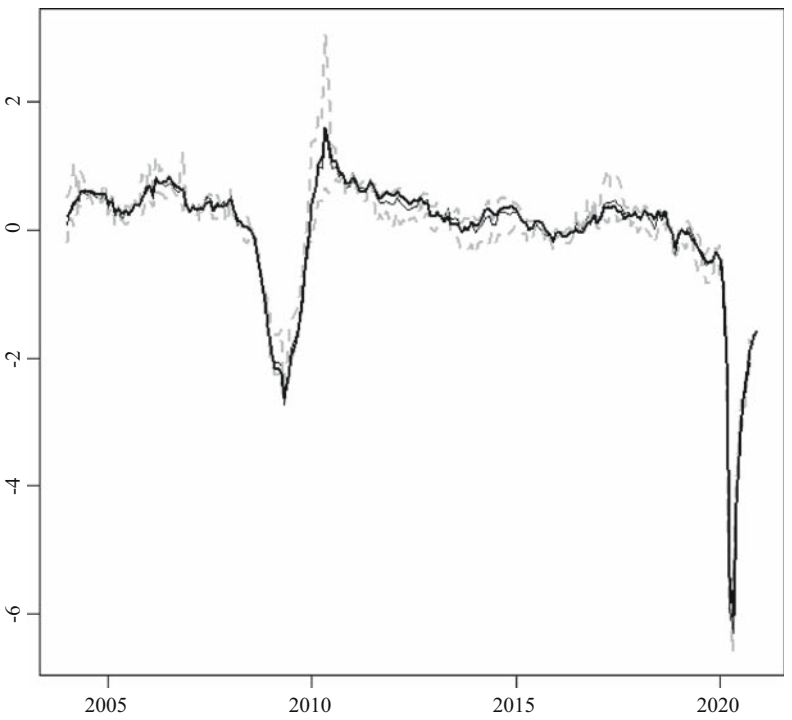



*Fig. 3. Factor estimates. The thin line is the static common factor, the gray dotted lines are their asymptotic confidence intervals at 95% and the solid line is the smoothed or dynamic common factor.*

contribution of each variable in the common factor. We compute the asymptotic confidence interval at 95% (Bai 2003) denoted by $CI_{\hat{P},0.05}$. Once the dynamic factor is estimated by using the Kalman smoother, it is necessary to reestimate the factor loadings to have $\hat{P} = f(\tilde{F})$, such that $\tilde{F} = g(\tilde{P})$. To do so, we use Monte Carlo estimation iterating 1,000 samples and select the replication that best satisfies the following condition:

$$\tilde{F} \approx X\tilde{P}/N \text{ s.t } \tilde{P} \in CI_{\hat{p},0.05}.$$

The results of the estimated factor loadings are shown in Figure 4. The loadings are ordered from the most positive contribution to the most negative.

We observe several similarities with respect to Figure 2. The more important variables in the factor estimates are (1) the industrial production, both from Mexico and the US, (2) imports, (3) hotel occupancy, (4) insureds by the Mexican Social Security System (IMSS), (5) exports and (6) the Google Trends topics such as "quarantine", "health crisis" and "face mask N95", which makes sense in the COVID-19 period. Obviously, when these variables are updated, it will be more important to update the nowcasts. Other variables, such as IMO, CONF MANUF, GAS, S&P 500, MOBILITY and E, are also relevant. The less important variables are M4 and remittances. Interestingly all variables are significant because their confidence intervals do not include the number zero.

We want to emphasize that the most relevant variables are also the timelier, for example the industrial production index of Mexico and the U.S. are updated around days ten and 16

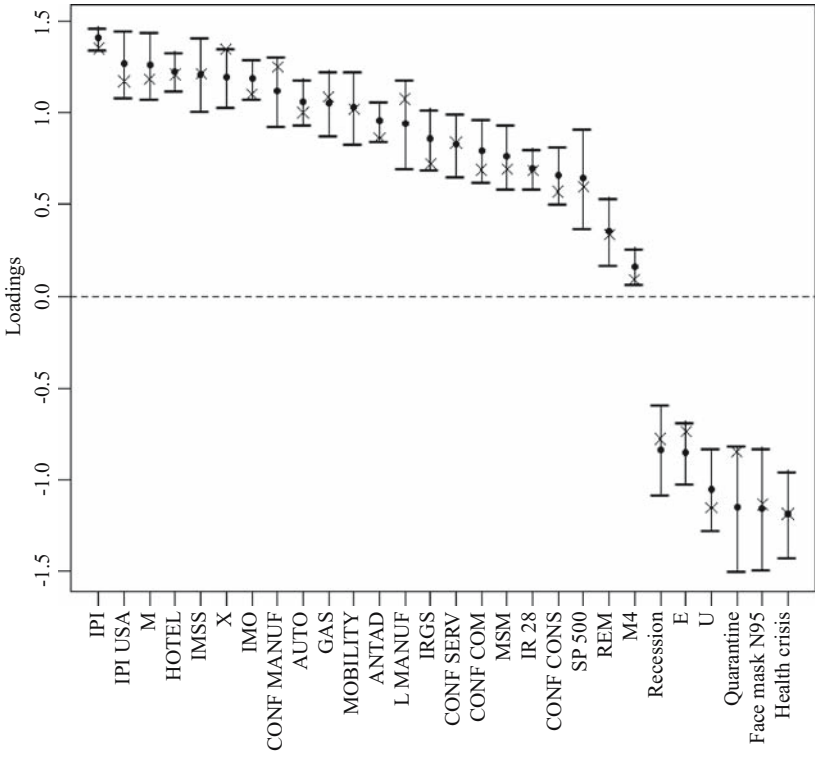



*Fig. 4. Factor loadings. The point is each $\hat{p}_i$ with its respective 95% asymptotic confidence interval. Crosses are the $\tilde{p}_i$.*

for $T^* + 1$ and $T^* + 2$, respectively, once closed the current month; furthermore, the exports and imports are updated for $T^* + 2$ by the 25th day, while IMO and IMSS are updated the first day and 12th day, respectively for $T^* + 2$. Consequently, this allows to have more accurate and correlated estimates beginning the first day of the current month for both, $T^* + 1$ and $T^* + 2$.

As we have previously noted, to obtain a consistent estimation of $\tilde{F}$ and $\hat{P}$ it is necessary that $\hat{\varepsilon}$ be stationary. We check this point with the PANIC test of Bai and Ng (2004). Table 1 summarizes the results.

We conclude that we have achieved stationarity in the idiosyncratic component, obtaining a statistic of 12.03 that generates a p-value of 0.00; hence, $\hat{\varepsilon}$ does not have a unit root. Additionally, we can verify with the augmented Dickey-Fuller test that $\tilde{F}$ is stationary with a p-value of 0.01; consequently, we have also achieved stationarity in $X$.

Note that we could have alternatively proposed to estimate the components of the DFMs in blocks, as suggested by Baríbura et al. (2011) by assuming that some common factors can be composed by different groups of variables, for example: macroeconomic, financial and non-traditional time series. However, given that the Onatski (2010) procedure detects one common factor, we considered that the assumption that all factors are global is adequate.

### 4.2. *Selecting the Nowcasts: Time Series Cross-Validation*

We apply the procedure described in Subsection 2.3 by using a $\Upsilon = (1/H_t, 1/H_t, \ldots, 1/H_t)'$; then, we assume that each AE has equal weight over time. Additionally, we fix $p_{\max} = q_{\max} = 6$. In empirical exercises, when $p_{\max}$ and $q_{\max} > 6$ the results do not change. Consequently, we assume that $p_{\max} = q_{\max} = 6$ is an appropiate window for searching models both from statistical and computational viewpoints. The obtained results indicate that the optimums $p*$ and $q*$ are selected to be equal to 2 with $k = 1$ in the factor equation. Consequently, the best model to carry out the current nowcasts is a regression model with ARMA(2,2) errors.

This specification generates the following nowcasts in CV one-step ahead, during $H_t = 36$ months that are presented in Figure 5.

We can see that the best nowcast model in CV performs well given that in 33 of 36 cases, 91.6%, the observed values are within the confidence interval at 95%. The MAE (equal weights in $\Upsilon$) is 0.52, and the mean absolute annual growth of IGAE is 3.50%. Regarding the median of the AEs, the estimated value is 0.38. Furthermore, based on Diebold-Mariano tests, 84 of 98 models are statistically equal to the best model using a significance of $\alpha = 0.05$, consequently, our nowcasting models generate consistently accurate results in the CV.

In addition, we compare our results to Corona et al. (2017a), which selects models in a similar way for CV period, we compare the IGAE levels' forecasts one-step ahead. They

*Table 1. PANIC test*

| Test | Value | P-value |
|---|---|---|
| ADF (dynamic factor) | -2.70 | 0.01 |
| Pooled test (idiosyncratic errors) | 12.03 | 0.00 |

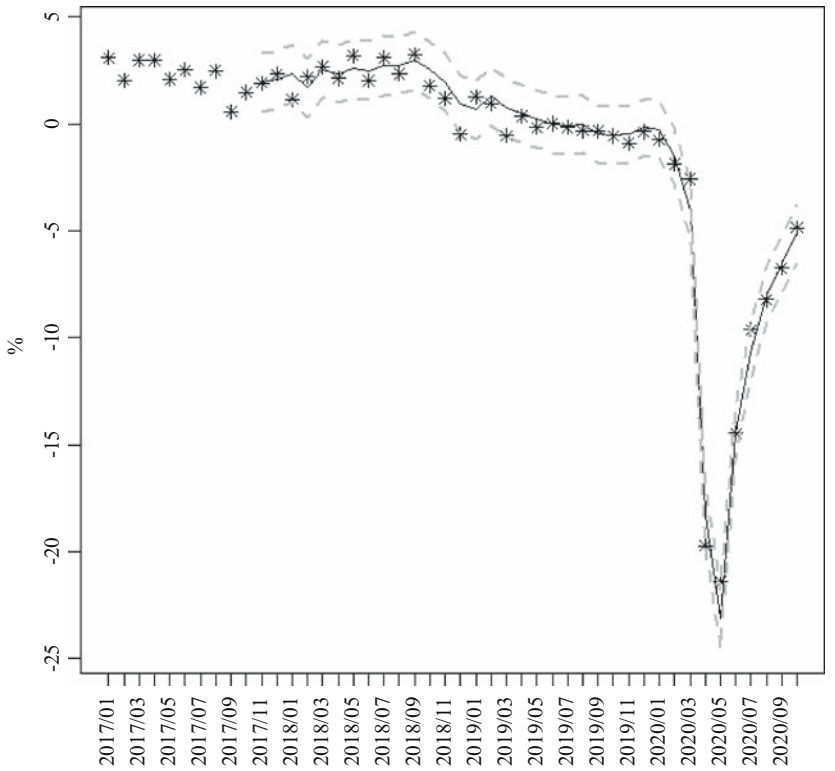



*Fig. 5. Nowcasts in CV. Asterisks are the observed values, the line depicts the nowcasts and the gray dotted lines are the confidence intervals at 95%.*

obtain a median of the root squared errors of around 0.4 and 0.5 in a period of 13 months (March 2015 to April 2016) while our model obtains a statistic of 0.43 for 36 months. Consequently, our approach tends to have better results.

To contrast the results of our approach with those obtained by other procedures in a certain period of CV, we consider the following two alternative models:

1. Static PC: We apply static PC to $X_t^*$ and by using the first estimated PC in a linear regression, we obtain the nowcasts for the last $H_t = 36$ months. In each nowcasting step we consider their specific relevant topics of Google Trends obtained by applying the procedure described in Subsection 2.1.1.
2. DFM without Google Trends: We estimate a traditional DFM similar to Corona et al. (2017a) or Gálvez-Soriano (2020), but using only economic and financial time series, that is, without considering the relevant topics extracted from Google Trends. Hence, we carry out the last $H_t = 36$ nowcasts.

Figure 6 shows the accumulated MAEs for the out-of-sample period by the previous two models and the obtained by the regression model with ARMA(2,2) errors.

We can see that, in cross-validation period the static PC is the one with the weakest performance, followed by the traditional DFM. Specifically, the MAE is 1.24 for the static PC, 0.74 when using DFM without Google Trends and, as we have commented, 0.52 for

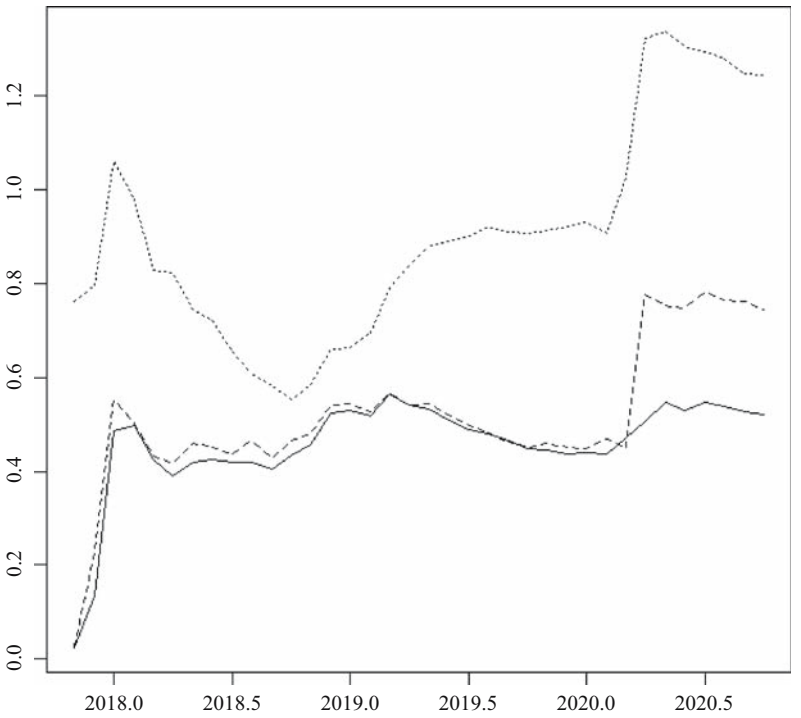



*Fig. 6. Cumulative MAEs for models for the period of CV. Solid line is the nowcasting approach suggested in this work, pointed line is the naive model and the dotted line is the traditional DFM (without Google trends topics).*

the incumbent model, which indeed includes this type of information. Note that the use of nontraditional information does not affect the behaviour of the MAEs previous to COVID-19 pandemic and reduces the error during this period. The Diebold-Mariano tests indicate that for all cross-validation period, we obtain $p$ values of 0.08 and 0.16 when comparing our approach with static PC and traditional DFM respectively, but since the pandemic period, reflected from April 2020, the MAEs of these models have increased from 1.02 to 1.24 for the static PC, and from 0.45 to 0.74 for the traditional DFM. For this period, the proposed DFM error is slightly increased from 0.47 to 0.52.

Note that the main reason why we cannot reject the null hypothesis can be attributed to the importance of the traditional time series, which are very relevant to timely estimate IGAE movements. Nonetheless, we can further reduce the error estimates during the shock attributed to the COVID-19 pandemic by using nontraditional information and, in the past, the error does not increase.

The conclusion is that the performance of the suggested approach is highly competitive when compared with (1) similar models for nowcasting of GDP, (2) models that estimate the levels of the objective variable and (3) alternative models that can be used in practice.

Although the residuals are not normally distributed, mainly due to the shocks introduced by the pandemic, note that residual normality is not required given that the likelihood estimation of nowcasting equations assumes asymptotic normality, consequently, the $t$ statistics are correctly estimated when no autocorrelation and no ARCH effect are present.

Instead, we do verify and validate the absence of autocorrelation and ARCH effect. Table 2 summarizes the results.

We can see that the residuals of the model with ARMA(2,2) errors does not have autocorrelation nor ARCH effect. Furthermore, the rest of the nowcasting models generate, on average, both non-autocorrelated and non-conditional-heteroskedastic residuals.

### *4.3. Current Nowcasts and Historical Analysis in Pseudo Real Time*

Having selected the highly competitive models in accordance with Diebold and Mariano (1995) tests, the final nowcasts for IGAE's annual percentage variation for November and December 2020 indicate a slight recovery of the economy in November and December 2020, obtaining nowcasts of -4.6% and -4.1%, respectively, with confidence intervals of (-6.3, -3.2) and (-5.6, -2.6) for both months. Considering the observed values for November and December, released on February 25 2021 by INEGI, the annual percentage change for the IGAE was -3.7% for both months; consequently, in this case the model generates nowcasts that falls within the confidence interval with deviations of -0.9% and -0.4% respectively.

The procedure described in the previous subsection allows the generation of nowcasts using databases with different cut off dates. In this way, we carry out the procedure updating the databases once a month during the last 35 periods. We assume similar conditions as in the current nowcasts, that is, we consider the same percentage of time series updated, 90% and 66% for $T^* + 1$ and $T^* + 2$ respectively. Consequently, we assume estimations in "pseudo real time" because they do not consider previous revisions in the databases. Additionally, the seasonal adjustment procedure tends to generate slight changes in the values of the time series in every update. However, note that the methods based on DFMs are sensible to distortions in the variance of the time series, consequently, it is preferable to estimate the components of the DFM with seasonal adjusted time series. Thus, we replicate the exercise with the current database assuming the last 35 months of its observed values. Note that this period encompasses several stages of the Mexican economy cycle: stability, slight drops, drastic drops and recovery. Note that transformations described in Subsection 2.1.2 are assumed fixed during all pseudo real time analysis. However, when generating an historical analysis about the optimal transformations along the last 37 months, only few variables present differences: business confidence (CONF CONS, CONF SERV and CONF COM), M4 and MOBILITY; which, reasonably, validates our assumption.

Figure 7 summarize the nowcasts results for $T^* + 1$ and $T^* + 2$ respectively, comparing them with the observed values.

*Table 2. Residual tests about the residuals of nowcasting models. We report the p-values.*

| Test | Model with ARMA(2,2) errors | Mean: All nowcasting models |
|---|---|---|
| Normality (Anderson-Darling test) | 0.02 | 0.05 |
| Autocorrelation (Ljung-Box test) | 0.80 | 0.86 |
| ARCH test (F statistic) | 0.29 | 0.30 |

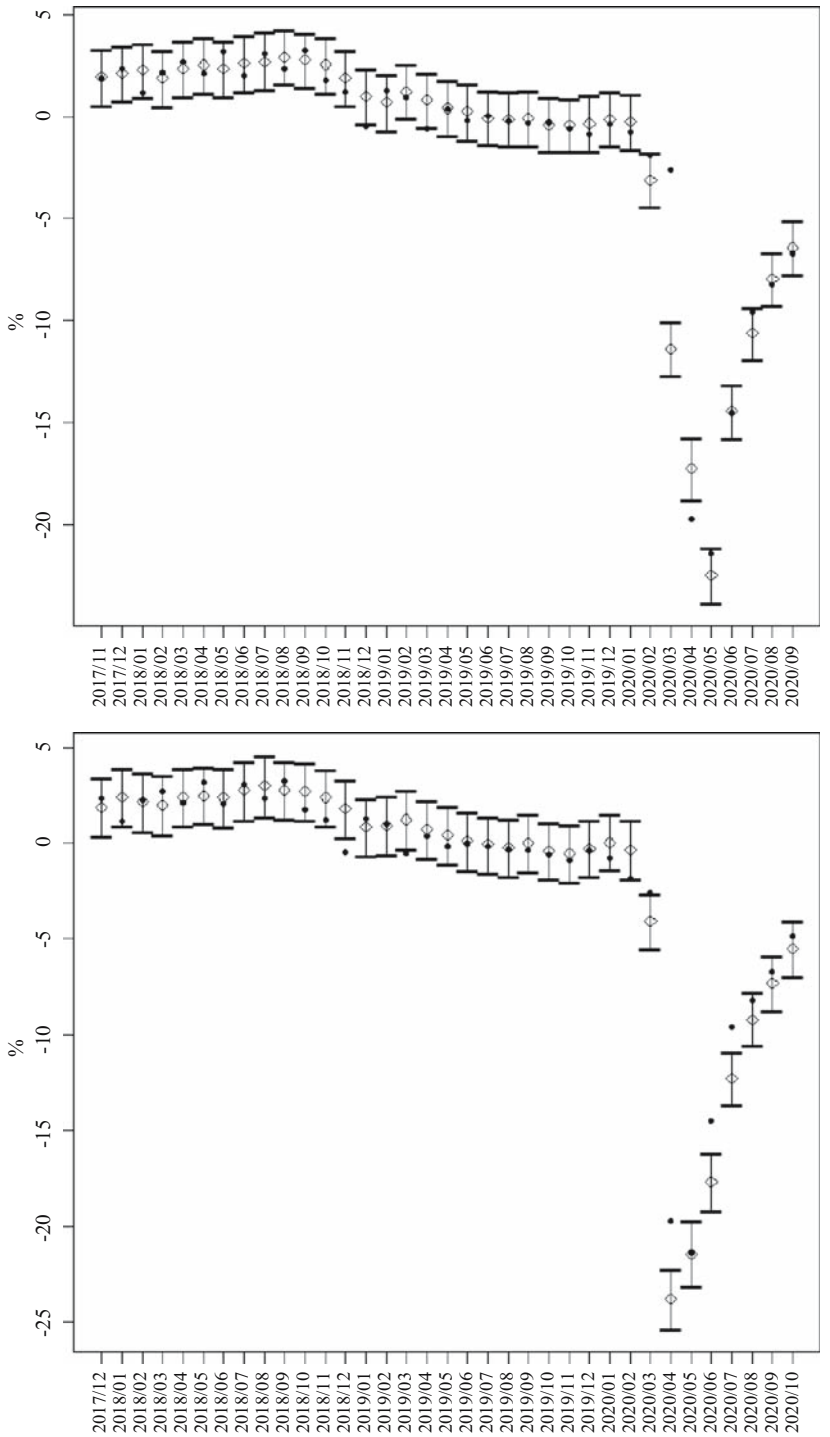



*Fig. 7. Nowcasts in pseudo real time for $T^*+1$ (top panel) and $T^*+2$ (bottom panel) during November 2017 to September 2020. The rhombus represent the nowcasts with their respective 95% confidence intervals. Points are the observed values.*

Focusing on $T^* + 1$, we can see the approach's functioning in the pre COVID-19 crisis period, and how on March 2020 it anticipated a larger fall than the observed value. The reason is that the selected Google Topics during these months overestimate the plunge, which was around -2.6% while the nowcasts for $T^* + 2$ was -4.1% and $T^* + 1$ of -11.4%. Note that in April of the same year, there was a historic fall, larger than 20%, which was outside our confidence interval. Nonetheless in all but these two months, the model has been precise in anticipating the falls and recoveries of the Mexican economy.

Specifically, for the March and April 2020 estimations, the dimension of the panels are $T = 182, N = 31$, and $T = 183, N = 30$, respectively. In turn, for March 2020 the relevant topics are: "coronavirus", "health crisis", "quarantine", "face mask", "face Mask N95", "dollar" and "recession". On the other hand, for April 2020 the selected Google Topics are: "coronavirus", "quarantine", "dollar", "face Mask N95", "pandemic" and "recession. Basic regression models, without assuming ARMA errors, generates in the $H_g^* = 8$ periods, MAEs of 0.96 and 1.05 for March and April 2020, respectively. Alternative procedures to LASSO regression as partial least squares select similar Google topics along the time.

Although Google topics are selected using the procedure described in subsection 2.1.1, they are drawn from a user-predefined list (see Appendix, Subsection 6.1). In this way, the accuracy of our procedure also depends on which topics are decided to start with. This list is used to carry out both, the current nowcasts (November and December 2020) and the previous 35 months. Thus, we acknowledge that this fact represents a limitation of our pseudo real time analysis given that the accuracy depends on which are the predefined topics along the time, above all, when unexpected events can occur.

Considering this fact as given, note that $T^* + 2$ shows consistent results with $T^* + 1$, and help us to conclude that the approach behaved well previous to the pandemic, but also during these phenomena. Note that, the model anticipated the drastic falls of April and May 2020, and subsequently it has showed precise performance, considering that these nowcasts are estimated with a smaller amount of information. In absolute terms, the MAE for $T^* + 1$ *y* $T^* + 2$ are 0.80 and 0.86 respectively, with AE's median of 0.42 and 0.58. Consequently, during several months the performance of the approach tends to generate accurate nowcasts, whether, the Mexican economy is stable or volatile.

A nowcasting model to compare with our approach is INEGI's "Early Monthly Estimation of Mexico's Manufacturing Production Level (INEGI. d.) whose target variable is manufacturing activity, generating the one step ahead nowcasts by using a timely electricity indicator. The average MAE for the annual percentage variation of manufacturing activity in the last 36 months, from November 2017 to October 2020, is 1.43. Consequently, in a similar sample period, we have a smaller average MAE than another nowcasting model where its monthly target variable is specified as annual percentage variation.

To compare the nowcasts/prediction errors with other procedures in the same sample, we estimate $T^* + 1$ and $T^* + 2$ using some benchmark univariate models, based on statistical methods such as AR. The AR models were obtained using the automatic models generated by the `auto.arima` function from `forecast R` package. and machine learning approaches based on neural networks (NN). Additionally, we consider DFMs without Google Trends (DFM-2). With respect to NN, we consider a multilayer perceptron (MLP) model and extreme learning machines (ELM), both adapted for time series

forecasting, see, for example, Huang et al. (2006) and Ord et al. (2017). NN are efficient non-parametric models which have been used in a wide variety of tasks, due to their ability to obtain useful data representations by means of non-linear activation functions. Although there are some NN architectures specialized in time series forecasting with remarkable results, see Gamboa (2017), Sezer et al. (2020), Hewamalage et al. (2021), among many others, they are very expensive in time and computational resources, mainly due to the high number of parameters involved. For this reason, we decided to use an ensemble of 20 single layer MLP and ELM, where the number of hidden nodes and other hyperparameters were selected automatically with a CV criterion, and the output layer weights for ELM are estimated with CV-LASSO. In both cases, we used the median as the combination operator for forecast. See Kourentzes et al. (2014) for details. We use the implementation of the `nnfor R` package. For all benchmark methods, we do not use exogenous regressors.

Table 3 shows the relative MAEs and the Diebold-Mariano p-values of the tests, both with respect to the DFM presented in this work.

We can see that, although the Diebold-Mariano test only rejects the null hypothesis of equal performance for the AR model, the other approaches generate larger MAEs than the DFM, above all, for $T^* + 2$. Consequently, our approach performs by generating lower estimation errors than some alternative procedures.

The inclusion of exogenous variables in NN methods generates more exact comparisons with respect to DFM and DFM-2. Thus, it is expectable that the errors may decrease by including these regressors.

Our approach generates monthly nowcasts of the Mexican economic activity, with results that are very close to the monthly GDP. Consequently, we carry out the quarterly version of GDP nowcasts by assuming that we know the IGAE for the first month of the quarter and then estimating the next two months ($T^* + 1$ and $T^* + 2$) with our pseudo real time analysis. We compare these quarterly nowcasts with the median of the Banco de Mexico Survey of Professional Forecasters (SPF) generated for the expected GDP annual variation of the corresponding quarter made after the first month ends. Table 4 shows the results for 2018:Q1 to 2020:Q3.

We can see that, our approach generated better estimates than the median of the expectations of SPF, which corroborates that our nowcasting approach is frequently much better with respect to the opinion of policy makers and specialized analysts. Specifically, the Diebold-Mariano test allows to conclude that our approach generates better nowcasts by obtaining a p-value of 0.04, that is, our method is more accurate than SPF estimates.

*Table 3. Relative MAEs and Diebold-Mariano p-values for $T^* + 1$ and $T^* + 2$ for each alternative method with respect to the DFM.*

| | MAE | | Diebold-Mariano test | |
|---|---|---|---|---|
| Model | $T^* + 1$ | $T^* + 2$ | $T^* + 1$ | $T^* + 2$ |
| AR | 3.127 | 3.316 | 0.06 | 0.02 |
| ELM | 1.926 | 2.842 | 0.17 | 0.13 |
| MLP | 1.864 | 2.753 | 0.17 | 0.13 |
| DFM-2 | 1.163 | 1.768 | 0.36 | 0.24 |

*Table 4. Comparison with Banxico's Survey of Proessional Forecasters:*

| | Percentage annual variation | | | Absolute errors | |
|---|---|---|---|---|---|
| Date | Observed GDP | SPF (median) | Nowcasting (quarterly version) | SPF (median) | Nowcasting (quarterly version) |
| 2018:Q1 | 2.50 | 1.80 | 1.82 | 0.70 | 0.68 |
| 2018:Q2 | 2.18 | 2.50 | 2.27 | 0.32 | 0.09 |
| 2018:Q3 | 2.81 | 2.45 | 2.94 | 0.36 | 0.13 |
| 2018:Q4 | 1.23 | 2.10 | 1.79 | 0.87 | 0.56 |
| 2019:Q1 | 0.26 | 1.50 | 1.25 | 1.24 | 0.99 |
| 2019:Q2 | 0.04 | 1.10 | 0.18 | 1.06 | 0.14 |
| 2019:Q3 | -0.12 | 0.60 | -0.13 | 0.72 | 0.01 |
| 2019:Q4 | -0.86 | 0.18 | -0.39 | 1.04 | 0.47 |
| 2020:Q1 | -1.98 | 0.54 | -2.56 | 2.52 | 0.58 |
| 2020:Q2 | -18.80 | -14.00 | -20.00 | 4.80 | 1.20 |
| 2020:Q3 | -8.65 | -12.00 | -8.40 | 3.35 | 0.25 |
| | | | Mean: | 1.54 | 0.46 |

## 5. Conclusions and Further Research

In this article, we contribute to the nowcasting literature by focusing on the one and two step-ahead of the annual percentage variation of IGAE, the equivalently of the Mexican monthly GDP, during the COVID-19 times. For this purpose, we use statistical and econometric tools to obtain accurate and timely estimates, some around 40 days before the publication of the official data. In summary, the suggested approach consists in (1) using the LASSO regression, along with empirical computational methods, to select the relevant topics that affect the IGAE in the short term that with traditional information allows for building a correlated and timely database to exploit the correlation among the variables and the IGAE, (2) estimate a dynamic factor by using the 2SM approach and (3) selecting linear regressions with ARMA errors in a cross-validation period to select the better models and generate the current nowcasts.

We highlight the following key results. We can see that our approach is highly competitive when compared with models such as static PC, AR, benchmark ELM and MLP; traditional DFM and other models published in the literature. Our procedure frequently captures the observed value, both, in cross-validation period and in real time for the time previous and during the period of the COVID-19 where the usage of Google Trends benefits in the reduction of the error estimation. Another contribution of this article lies in its statistical point of view, given that we compute the confidence interval of the factor loadings and the factor estimates, verifying the significance of the factor on each variable and the uncertainty attributed to the factor estimates. Additionally, we consider some econometric issues to guarantee the consistency of estimates like stationarity in idiosyncratic noises and uncorrelated errors in nowcasting models. Finally, it is of interest to note out-of-sample performance, whether the nowcast error increases when using monthly versus quarterly data.

We highlight the relevance of this method to produce official statistics. Our approach generates preliminary and accurate estimates of the IGAE, a variable that is closely pegged

to GDP. This allows the policy makers and market participants to make informed decisions about the near future of the economy.

Future research topics emerged when doing this research. The most important is to incorporate machine learning techniques to design an algorithm that automatically select the set of possible relevant topics from Google Trends, also explore the inclusion of news and other nontraditional and unstructured information different to predefined lists. Other, is the implementation of an algorithm to estimate non-stationary common factors and make the selection of the number of factors flexible, such as the one developed in Corona et al. (2020), to minimize a measure of nowcasting errors. Also, it would be interesting to incorporate IPI information as restrictions to the nowcasts, by exploring some techniques that incorporate nowcasts restrictions when official countable information is available. We must note that, we can introduce electronic payment information provided by Bank of Mexico in a similar way as Galbraith and Tkacz (2018) and León and Ortega (2018) and compare the results with a large scale DFM. Also, it is interesting the possibility to include time varying intercept as Antolin-Diaz et al. (2017) and González-Astudillo and Baquero (2019) and incorporate dynamic factors as exogenous regressors in machine learning methods is interesting. Furthermore, for future research in this area, it is worth to study further the effects of monthly timely estimate variables versus quarterly time series in nowcasting models; this can be achieved by Monte Carlo analysis with different data generating processes which can occur in practice to compare the increase in the error estimation when distinct frequencies and transformations of time series are used. Finally, it is of interest to generate different transformations for each $h$ during the pseudo real time analysis and to test LASSO or elastic net regression as variable selection method for both traditional and nontraditional blocks of time series.

## 6. Appendix

### *6.1. Database*

Traditional macroeconomic time series

| Short | Variable | Source | Time Span |
|---|---|---|---|
| ANTAD | Total sales of departmental stores | ANTAD | 2004/01-2020/11 |
| AUTO | Automobiles production | INEGI | 2004/01-2020/12 |
| CONF COM | Right time to invest (Commerce) | INEGI | 2011/06-2020/12 |
| CONF CONS | Right time to invest (Construction) | INEGI | 2011/06-2020/12 |
| CONF MANU | Right time to invest (Manufacturing) | INEGI | 2004/01-2020/12 |
| CONF SERV | Right time to invest (Services) | INEGI | 2017/01-2020/12 |
| GAS | Fuel demand | SENER | 2004/01-2020/12 |
| HOTEL | Hotel occupancy | Tourism secretariat | 2004/01-2020/11 |
| IMO | Index of manufacturing orders | INEGI | 2004/01-2020/12 |
| IMSS | Permanent and eventual insureds to the Social Security | IMSS | 2004/01-2020/12 |
| IPI | Industrial Production Index | INEGI | 2004/01-2020/11 |
| IPI USA | Industrial Production Index (USA) | BEA | 2004/01-2020/12 |
| IRGS | Income of retail goods and services | INEGI | 2008/01-2020/10 |
| L MANUF | Trend of labor in manufacturing | INEGI | 2007/01-2020/10 |
| M | Total imports | INEGI | 2004/01-2020/11 |
| M4 | Monetary aggregate M4 | Banxico | 2004/01-2020/11 |
| REM | Total remittances | Banxico | 2004/01-2020/11 |

| Traditional macroeconomic time series | | | |
|---|---|---|---|
| **Short** | **Variable** | **Source** | **Time Span** |
| U | Unemployment rate | INEGI | 2005/01-2020/11 |
| X | Total exports | INEGI | 2004/01-2020/11 |
| **Traditional financial time series** | | | |
| **Short** | **Variable** | **Source** | **Time Span** |
| E | Nominal exchange rate | Banxico | 2004/01-2020/12 |
| IR 28 | Interest rate (28 days) | Banxico | 2004/01-2020/12 |
| MSM | Mexican stock market index | Banxico | 2004/01-2020/12 |
| SP 500 | Standard & Poor's 500 | Yahoo! finance | 2004/01-2020/12 |
| **Nontraditional time series** | | | |
| **Short** | **Variable** | **Source** | **Time Span** |
| AH1N1 | AH1N1 online search index | Google | 2004/01-2020/12 |
| AMLO | AMLO online search index | Google | 2004/01-2020/12 |
| Ayotzinapa | Ayotzinapa online search index | Google | 2004/01-2020/12 |
| Calderon | Calderon online search index | Google | 2004/01-2020/12 |
| Cartel | Cartel online search index | Google | 2004/01-2020/12 |
| Casa Blanca | Casa Blanca online search index | Google | 2004/01-2020/12 |
| Chapo | Chapo online search index | Google | 2004/01-2020/12 |
| China | China online search index | Google | 2004/01-2020/12 |
| Coronavirus | Coronavirus online search index | Google | 2004/01-2020/12 |
| Corrupcion | Corrupcion online search index | Google | 2004/01-2020/12 |
| Crisis economica | Crisis economica online search index | Google | 2004/01-2020/12 |
| Crisis sanitaria | Crisis sanitaria online search index | Google | 2004/01-2020/12 |
| Cuarentena | Cuarentena online search index | Google | 2004/01-2020/12 |
| Cubrebocas | Cubrebocas online search index | Google | 2004/01-2020/12 |
| Desempleo | Desempleo online search index | Google | 2004/01-2020/12 |
| Dólar | Dolar online search index | Google | 2004/01-2020/12 |
| Elecciones | Elecciones online search index | Google | 2004/01-2020/12 |
| EPN | EPN online search index | Google | 2004/01-2020/12 |
| Gasolina | Gasolina online search index | Google | 2004/01-2020/12 |
| Homicidios | Homicidios online search index | Google | 2004/01-2020/12 |
| Huachicol | Huachicol online search index | Google | 2004/01-2020/12 |
| Inflacion | Inflacion online search index | Google | 2004/01-2020/12 |
| Inseguridad | Inseguridad online search index | Google | 2004/01-2020/12 |
| Mascarilla N95 | Mascarilla N95 online search index | Google | 2004/01-2020/12 |
| Medidas economicas | Medidas econoomicas online search index | Google | 2004/01-2020/12 |
| Migracion | Migracioon online search index | Google | 2004/01-2020/12 |
| Migrantes | Migrantes online search index | Google | 2004/01-2020/12 |
| MOBILITY | Media mobility index | Twitter | 2004/01-2020/12 |
| Morena | Morena online search index | Google | 2004/01-2020/12 |
| Muertos | Muertos online search index | Google | 2004/01-2020/12 |
| Muro | Muro online search index | Google | 2004/01-2020/12 |
| Pacto | Pacto online search index | Google | 2004/01-2020/12 |
| PAN | PAN online search index | Google | 2004/01-2020/12 |
| Pandemia | Pandemia online search index | Google | 2004/01-2020/12 |
| PEMEX | PEMEX online search index | Google | 2004/01-2020/12 |
| Peso | Peso online search index | Google | 2004/01-2020/12 |
| Petroleo | Petrooleo online search index | Google | 2004/01-2020/12 |
| PRI | PRI online search index | Google | 2004/01-2020/12 |
| Recesion | Recesioon online search index | Google | 2004/01-2020/12 |
| Reformas | Reformas online search index | Google | 2004/01-2020/12 |
| Salario | Salario online search index | Google | 2004/01-2020/12 |
| Sismo | Sismo online search index | Google | 2004/01-2020/12 |
| Tipo de cambio | Tipo de cambio online search index | Google | 2004/01-2020/12 |
| Trump | Trump online search index | Google | 2004/01-2020/12 |
| Violencia | Violencia online search index | Google | 2004/01-2020/12 |

### 6.2. Tables 1–3

*Table 1. Dotted lines represent the specifc Xi and solid lines indicate the specified y*. Numbers in parentheses indicate the linear correlation and those between brackets the transformation.*

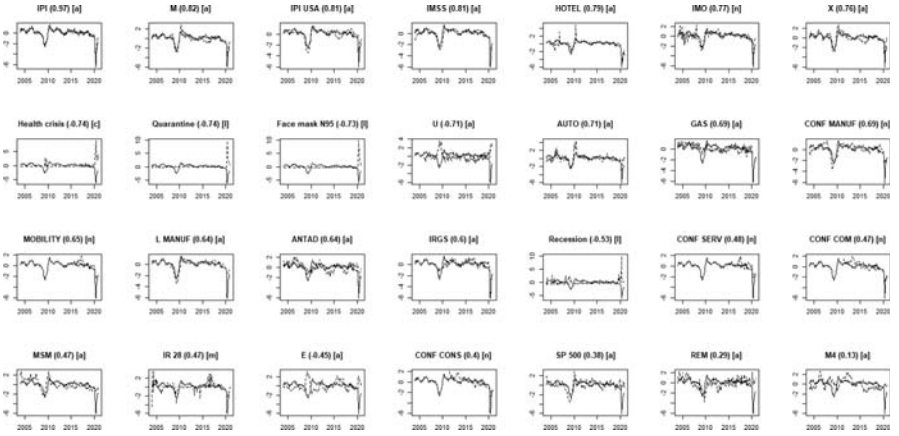



*Table 2. Percentage of updated information to carry out the nowcasts T* + 2 once the current month T* is closed.*

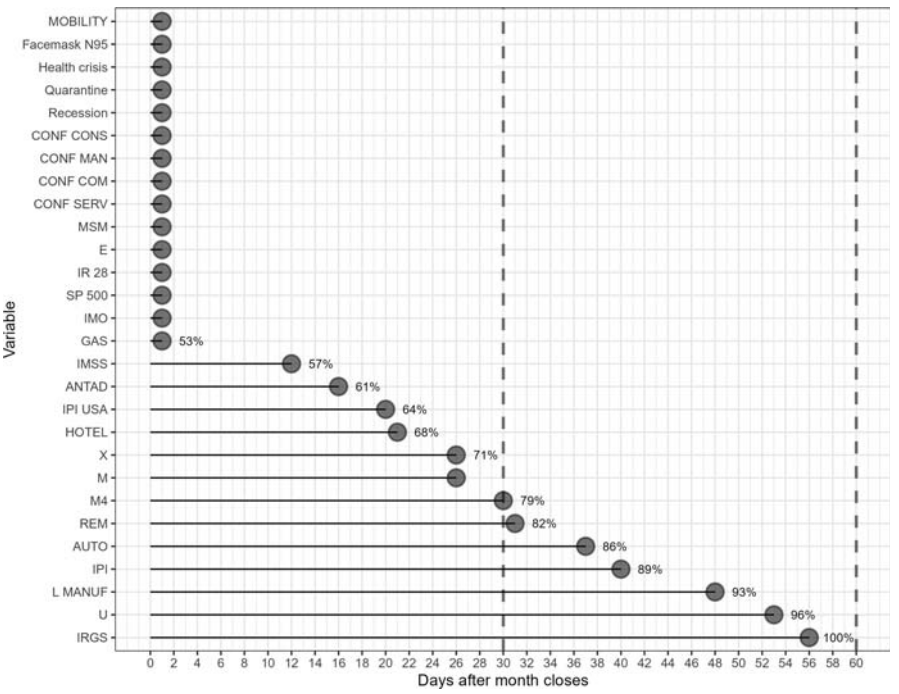
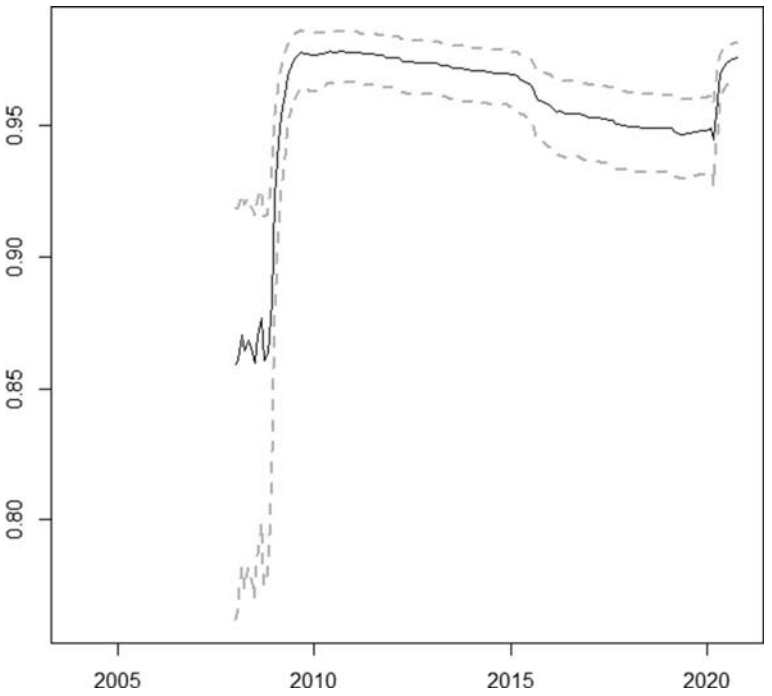


*Table 3. Blue line is Corr ($\tilde{F}_t$; y*) from January 2008 to October 2020. Lines represent the confidence interval at 95%.*




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